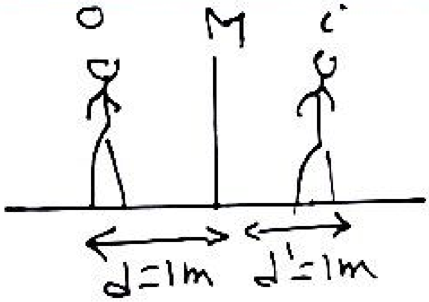
Set # (7)

2nd 2024

PHY102A Solution MKQ P1

24.1 $d' = 1m$



$$\text{distance} = d + d'$$
$$= 1 + 1 = \boxed{2m}.$$

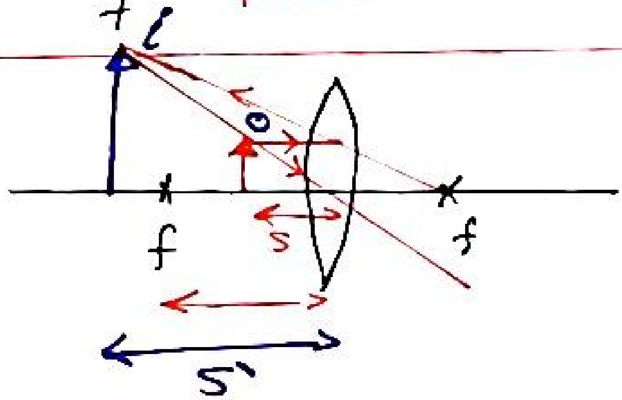
24.4 $R_1 = -0.5m$ concave

$R_2 = +0.7m$ convex

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$
$$= (1.6-1)\left(\frac{1}{-.5} + \frac{1}{0.7}\right) = 0.6\left(-2 + \frac{10}{7}\right)$$
$$= 0.6\left(\frac{-4}{7}\right) = -\frac{2.4}{7} = \boxed{-2.9m}$$

24.10 $f = +0.20m$

$s = 0.08m$



(b)

(a) $$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$$

$$\frac{1}{0.2} = \frac{1}{0.08} + \frac{1}{s'}$$

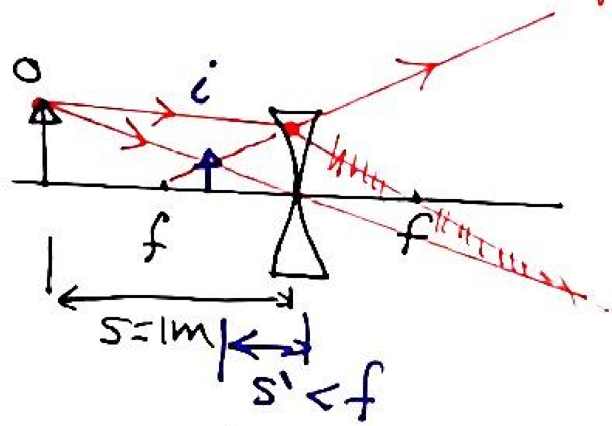
$$\frac{1}{s'} = \frac{1}{0.2} - \frac{1}{0.08} = \frac{0.08-0.2}{0.2 \times 0.08} = \frac{-0.12}{0.016} \Rightarrow s' = \frac{-1.6}{12} = \boxed{-1.33m}$$

(c) $$m = -\frac{s'}{s} = -\frac{-1.33}{0.08} = \boxed{1.67}$$

P2

24.11 $s = 1m = 2f$

$f = -0.5m$ ← concave



$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$$

$$\frac{1}{-.5} = \frac{1}{1} + \frac{1}{s'} \Rightarrow -2-1 = \frac{1}{s'}$$

$$\frac{1}{s'} = -3 \Rightarrow s' = \frac{-1}{3} = \boxed{-.33\,m}$$

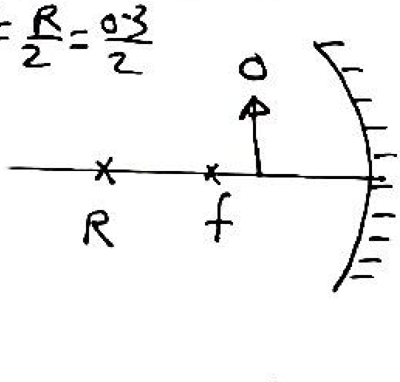
24.18 $P_1 = \frac{1}{f_1} = -8 \Rightarrow f_1 = \frac{-1}{8} = \boxed{-0.125m}$

$P_2 = \frac{1}{f_2} = -\cancel{8}6 \Rightarrow f_2 = \frac{-1}{6} = \boxed{-0.167m}$

24.19 $P = 4 \Rightarrow f = \frac{1}{4} = \boxed{0.25m}$

24.53 NOTE: $R = 0.3m \rightarrow f = \frac{R}{2} = \frac{0.3}{2}$

$f = 0.15m$



$$m = -\frac{s'}{s} = 3$$

$$s' = -3s$$

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'} = \frac{1}{s} - \frac{1}{3s} = \frac{3-1}{3s} = \frac{2}{3s}$$

$$\Rightarrow s = \frac{2}{3}f = \frac{R}{3} = \frac{0.3}{3} = \boxed{0.1m}.$$